Failure mode measured on the VCS	VCS measured evidence proven in this benchmark		Neural-network analogue documented, not proven here	Status here → there
1 **Gradient vanishes on discrete / index outputs**	**0.000** position-regime F single record	Vanilla saliency & IG faithfulness = 0.000 on the position regime; whole gradient/correlational bucket = 0.068 (n=9). Provably 0 at a strobe-timed argmax readout (§3.2 oracle).	Shattered / saturated gradients; non-differentiable argmax & index readouts (analogue refs G1, G2).	**proven here** **analogue there**
2 **Saliency is model-invariant (sanity-check fails)**	**0.000** sanity-check margin single record	Program-randomization test: saliency cosine = 1.000 to the real map while 95% of RAM changed -> sanity check FAILS. Constant Jacobian on the content path, no ReLU.	Saliency unchanged under model- & label-randomization; behaves like an edge detector (analogue ref S1).	**proven here** **analogue there**
3 **Correlation ≠ causation (Granger & tuning trap)**	**0.136** Granger F 95% CI [0.02, 0.29]	Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116 (spurious) -- vs the exact §3.2 oracle.	Spurious Granger edges; 'present ≠ used' tuning & attention-as-explanation (analogue refs C1, C2).	**proven here** **analogue there**
4 **Single-unit ablation has an interaction blind-spot**	**0.414** connectome F 95% CI [0.08, 0.78]	Single-unit lesion importance is faithful (0.987) yet MISSES wiring: the connectome graph is recovered at only 0.414; SAE ablations move y in 0 of the audited cells.	Single-unit ablation misleading; computation distributed across units (analogue refs D1, D2).	**proven here** **analogue there**
5 **SAE/probe: present-not-used**	**0.162** selectivity F 95% CI [0.08, 0.24]	SAEs match a variable (100% of candidates in 4/6 games, min 83%) yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are decodable-not-causal -> selectivity F = 0.162.	Control-task selectivity gap; superposition / polysemantic neurons (analogue refs P1, P2).	**proven here** **analogue there**
6 **IG/EG baseline sensitivity**	**0.034** EG F (all-regime) 95% CI [0.00, 0.10]	IG magnitude is baseline-sensitive (corr. sens. up to 0.999 across games); a baseline = the content byte kills IG entirely; Expected Gradients lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input (analogue refs B1, B2).	**proven here** **analogue there**

Analogue references (documented in the literature, not measured here):
G1 Balduzzi et al. 2017; G2 Sundararajan et al. 2017 (IG). S1 Adebayo et al. 2018. C1 Anand et al. 2019; C2 Jain & Wallace 2019.
D1 Morcos et al. 2018; D2 Leavitt & Morcos 2020. P1 Hewitt & Liang 2019; P2 Elhage et al. 2022. B1 Kindermans et al. 2019; B2 Sturmfels et al. 2020.

All VCS numbers read from committed records (leaderboard.json, leaderboard_ci.csv, faithful_demo.json, guided_backprop / ig_baseline_sweep / sae / linear_probing core summaries); 95% CIs are bootstrap-over-games (P2-R-U